A. Course Handout (Version 1.0)

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Cloud Computing	Session	2025-2026
Course Code	23CS009	Semester/Batch	5th/2023
L-T-P (Per Week)	2-0-0	Course Credits	04
Pre-requisite	Basic concepts of Programming, Storage, Databases	NHEQF Level	5.5
Course Coordinator	Dr. Priyanka Gupta	SDG Number	4,8,9

CLO01	Understand the fundamental concepts and market perspective of cloud computing
CLO02	Gain familiarity with various AWS tools and services.
CLO03	Perform hands-on practice on the AWS platform.
CLO04	Effectively manage, upscale, and downscale a business using AWS.
CLO05	Design and deploy secure and scalable cloud-based solutions.

1. Objectives of the Course

This course will introduce Cloud computing to attendees from basic concept and will take them to hand-on journey. The main objectives of the course are:

- Broad overview of cloud platforms and difference between public, private, hybrid cloud.
- How to start on cloud platform and Learn foundation of respective cloud most common infrastructure instance i.e., AWS.
- Hands on practice with AWS platform for high level of customization to meet the requirement of individual businesses.
- How AWS allows users to effortlessly monitor and handle specific resources.
- How AWS tags can be utilized for cost tracking, security and automation as well.

2. Course Learning Outcomes

After completion of the course, student should be able to:

	Course Learning Outcome	*Pos	**CL	***KC	Sessions
CLO01	Understand the fundamental concepts and market perspective of cloud computing.	PO1,PO3, PO4, PO8, PO12	K2	Factual Conceptual	**15**
CLO02	Gain familiarity with various AWS tools and services.	PO1,PO3, PO4,PO5, PO8, PO12	K3	Conceptual Procedural	**23**
CLO03	Perform hands-on practice on the AWS platform.	PO1, PO2, PO3, PO4,PO5,PO6, PO8,P10, PO11, PO12	K3	Conceptual Procedural	**32**

CLO04	Effectively manage, upscale, and downscale a business using AWS.	PO1, PO2, PO3, PO4,PO5,PO7, PO8,PO10, PO11, PO12	K4	Conceptual Procedural	**24**
CLO05	Design and deploy secure and scalable cloud-based solutions.	PO1, PO2, PO3, PO4,PO6,PO8, PO9, PO11, PO12	K3	Conceptual Procedural	**18**
Total Contact Hours					**112**

Revised Bloom's Taxonomy Terminology

* PO's available at (shorturl.at/cryzF)

**Cognitive Level =CL

***Knowledge Categories = KC

Course Learning Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CLO01	H	H	H	M					L		L	H
CLO02	H	H	H	M					L		L	H
CLO03	H	H	H	H					M		M	H
CLO04	H	H	H	H	M				H		M	H
CLO05	H	H	H	H	M				H		M	H

H=High, M=Medium, L=Low

3. ERISE Grid Mapping

Feature Enablement	**Level (1-5, 5 being highest)**
Entrepreneurship	4
Research	3
Innovation	5
Skills	5
Employability	5

4. Recommended Books (Reference Books/Textbooks)

B01: Architecting the Cloud: Design Decisions for Cloud Computing Service Models by Michael J. Kavis, 2014.
BO2: AWS The Ultimate Guide to Amazon Web Services Step-by-step Guide from Beginners to Advanced! by Stephen Springwell, Amplitude Ltd., Feb-2021.
BO3: Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud, First Edition by Pearson (Mark Wilkins), Nov-2019.

5. Other readings & relevant websites

S.N.	Link of Journals, Magazines, Websites, and Research Papers
1	https://computingforgeeks.com/how-to-learn-cloud-computing-from-beginner-to-expert/#:~:text=Learn%20Cloud%20Computing%20from%20Scratch%201%20Step%20One%3A,Six%3A%20Get%20Deep%20in%20your%20area%20of%20expertise
2	https://aws.amazon.com/free/database/?trk=ce743268-ef4c-43aa-9909-188edfb237f9&sc_channel=ps&s_kwcid=AL!4422!10!71331043936464!71331563897761&s_kwcid=AL!4422!10!71331043936464!71331563897761&ef_id=3129036f1e2f156b980c608019841b12:G:s
3	https://www.simplilearn.com/cloud-solutions-architect-masters-program-training?utm_source=bing&utm_medium=cpc&utm_term=simplilearn%20AWS%20training&utm_content=369201188-1329310345774847-&utm_device=c&utm_campaign=B-Search-Brand-Broad-IN-AllDevice-adgroup-Brand-Cat-AWS&msclkid=27c934140d0110e2801257fb0aaa501c

6. Recommended Tools and Platforms

AWS - https://aws.amazon.com/

7. Course Plan

Lecture Number	Topics	Recommended Books
1-8	Cloud Computing Basics – History of Cloud Computing, Characteristics of Cloud Computing, Need for Cloud computing, Advantages and Possible Disadvantages of cloud computing, Cloud Computing Architecture, Real-World Applications of Cloud Computing.	BO1, BO2
9-15	Introduction to Cloud Deployment Models, Types of Cloud Deployment Models: Public, Private, Hybrid and Community, Comparison: Public vs. Private vs. Hybrid Cloud. Cloud Service Models- Introduction to Cloud Service Models, Infrastructure as a Service (IaaS) Layer, Platform as a Service (PaaS) Layer, Software as a Service (SaaS), Difference Between IaaS, PaaS, and SaaS, Cloud data lifecycle phases.	BO3
16-22	Cloud service providers- Overview and history of Amazon Web Services, Introduction to EC2, Instance Types and Uses, Auto scaling Instances, Introduction to Amazon Machine Images (AMI's), key components and types of AMI's, Elastic IPs, Elastic Load Balancing.	BO3
23-30	Cloud storage- Introduction to Amazon S3 Storage, Buckets and Objects in S3, S3 Storage Classes, Amazon EBS (Elastic Block Store), Amazon Database Services: Amazon RDS (Relational Database Service), Amazon DynamoDB, Amazon Redshift etc.	BO3
	ST-1 (Syllabus covered from Lecture 1 to 30)	
31-34	AWS Networking Services- Introduction to AWS Networking, Introduction to VPN and VPC	BO2
35-40	AWS CloudTrail and CloudWatch and alarms, Docker, Kubernetes, and Elastic Beanstalk, CloudWatch, AWS Messaging services (SNS, SQS, SES) and Data Security.	BO2

41-48	Legal and Economic Challenges in Data Protection- Data Recovery and protection, Backup Strategies and recovery considerations, Copyright and Intellectual Property Rights (IPR), International Legal Frameworks.	BO3
49-60	Budget and Cost Management- Budgeting, Resource planning, Introduction to ML ecosystem, Amazon Comprehend, Amazon Forecast, Amazon Fraud Detector, Amazon Kendra, Amazon Lex, Amazon Sage Maker, Amazon Textract, Amazon Transcribe, Amazon Translate.	BO2
	ST-2 (Syllabus covered from Lecture 31 to 60)	
END TERM – FULL SYLLABUS		

8. Delivery/Instructional Resources

Lecture No.	Topics	Web References	Audio-Video
1-8	Cloud Computing Basics – History of Cloud Computing, Characteristics of Cloud Computing, Need for Cloud computing, Advantages and Possible Disadvantages of cloud computing, Cloud Computing Architecture, Real-World Applications of Cloud Computing.	https://aws.amazon.com/what-is-cloud-computing/	https://youtu.be/ZHCtVZ6cjdg
9-15	Introduction to Cloud Deployment Models, Types of Cloud Deployment Models: Public, Private, Hybrid and Community, Comparison: Public vs. Private vs. Hybrid Cloud. Cloud Service Models- Introduction to Cloud Service Models, Infrastructure as a Service (IaaS) Layer, Platform as a Service (PaaS) Layer, Software as a Service (SaaS), Difference Between IaaS, PaaS, and SaaS, Cloud data lifecycle phases.	https://aws.amazon.com/types-of-cloud-computing/	https://youtu.be/fZ3D6HQrWzs
16-22	Cloud service providers- Overview and history of Amazon Web Services, Introduction to EC2, Instance Types and Uses, Auto scaling Instances, Introduction to Amazon Machine Images (AMI's), key components and types of AMI's, Elastic IPs, Elastic Load Balancing.	https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html	https://www.youtube.com/watch?v=8TlukLu11Yo

23-30	Cloud storage- Introduction to Amazon S3 Storage, Buckets and Objects in S3, S3 Storage Classes, Amazon EBS (Elastic Block Store),Amazon Database Services: Amazon RDS (Relational Database Service), Amazon DynamoDB, Amazon Redshift etc.	https://docs.aws.amazon.com/whitepapers/latest/aws-overview/database.html https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html	https://youtu.be/oaZ3R4NCRu8
31-34	AWS Networking Services- Introduction to AWS Networking, Introduction to VPN and VPC	https://docs.aws.amazon.com/vpn/latest/s2svpn/SetUpVPNConnections.html	https://youtu.be/b1b6JTYnbjU
35-40	AWS CloudTrail and CloudWatch and alarms, Docker, Kubernetes, and Elastic Beanstalk, CloudWatch, AWS Messaging services (SNS, SQS, SES) and Data Security.	https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/WhatIsCloudWatch.html	https://www.youtube.com/watch?v=qVYnlxdEebE
41-48	Legal and Economic Challenges in Data Protection- Data Recovery and protection, Backup Strategies and recovery considerations, Copyright and Intellectual Property Rights (IPR), International Legal Frameworks.	https://aws.amazon.com/compliance/data-protection/ https://aws.amazon.com/backup-restore/	https://surl.li/qaacjj
49-60	Budget and Cost Management- Budgeting, Resource planning, Introduction to ML ecosystem, Amazon Comprehend, Amazon Forecast, Amazon Fraud Detector, Amazon Kendra, Amazon Lex, Amazon Sage Maker, Amazon Textract, Amazon Transcribe, Amazon Translate.	https://www.techtarget.com/searchdisasterrecovery/tip/Business-continuity-in-the-cloud-Benefits-and-planning-tips https://aws.amazon.com/sagemaker/	https://youtu.be/LcAPj95KeSA https://youtu.be/Imj8L-po4kw

9. Lab Plan

Lab Number	Practical	Learning Resources
1-8	Sign up for AWS, explore free-tier services, and launch EC2 instances using Ubuntu/Linux and Windows operating systems.	https://www.youtube.com/watch?v=xiZ5MyVX8ro

9-15	Connect to an EC2 instance via EC2 Connect, SSH Client, AWS CLI, and PuTTY SSH Client using a key pair file. Access and modify Security Group inbound and outbound rules to control traffic for EC2 instances.	YouTube: "How to Connect to EC2 Using PuTTY and Key Pairs" by Simplilearn.
16-22	Set up and configure web servers on EC2 instances, including Apache and Nginxserver. Configure auto-scaling for EC2 instances to handle variable traffic.	AWS Hands-on for Beginners by Coursera. Auto Scaling EC2 Instances Guide.
23-30	Create an Amazon S3 bucket, upload objects, set up access controls with bucket policies, and use it for hosting static content.	YouTube: "How to Host a Static Website on S3" by Simplilearn.
31-34	Set up an AWS Elastic Load Balancer (ELB) to distribute traffic among EC2 instances.	Elastic Load Balancer in AWS by Linux Academy.
35-40	Set up Amazon VPC networking, including subnets, route tables, security groups. Scenario 1:VPC With a Public Subnet Only (Standalone Web) Scenario 2: VPC with Public and Private Subnets (3 Tier App)	AWSVPC Basics. https://www.youtube.com/watch?v=3K19Dzyrlq8
41-48	Configure Amazon CloudWatch to monitor EC2 instance metrics and set up alarms.	AWS CloudWatch Metrics Tutorial. CloudWatch Monitoring and Alarms
49-60	Create an SNS, subscribe and publish a message, set up an SQS queue, integrate it with an EC2 instance, and use SES to send event notifications via email.	https://docs.aws.amazon.com/sns/latest/dg/welcome.html

10. Action plan for different types of learners

Slow Learners	Average Learners	Fast Learners
• Remedial Classes • Encouragement for improvement using Peer Tutoring • Use of Audio and Visual Materials • Use of Real-Life Examples	• Expert Lecture • Formative Exercises used to highlight concepts and notions • E-notes and E-exercises to read ahead of the pedagogic material.	• Engaging students to hold hands of slow learners by creating a Peer Tutoring Group • Design solutions for complex problems • Design solutions for complex problems • Presentation on topics beyond those covered in CHO

11. Evaluation Scheme & Components

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment
Component 1	Continuous Evaluation(CE)/Lab assessment	02	20%	Offline
Component 2	Sessional Tests (STs)	02*	30%	Offline
Component 3	End Term Examination	01	50%	Offline
Total		**100%**		

* Out of 02 STs, the score of Best 1 ST will be considered.

* Practical File submission is mandatory and will be considered for internal component.

* Makeup Examination will compensate for either ST-1 or ST-2 (Only for genuine cases, based on the Dean's approval and the exam will be based on the full syllabus).

**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

12. Syllabus of the Course

Subject: Cloud Computing	**Subject Code:** 22CS035	
Contents	**Lectures**	**Weightage (%)**
Cloud Computing Basics – History of Cloud Computing, Characteristics of Cloud Computing, Need for Cloud computing, Advantages and Possible Disadvantages of cloud computing, Cloud Computing Architecture, Real-World Applications of Cloud Computing.	8	15%
Introduction to Cloud Deployment Models, Types of Cloud Deployment Models: Public, Private, Hybrid and Community, Comparison: Public vs. Private vs. Hybrid Cloud. Cloud Service Models- Introduction to Cloud Service Models, Infrastructure as a Service (IaaS) Layer, Platform as a Service (PaaS) Layer, Software as a Service (SaaS), Difference Between IaaS, PaaS, and SaaS, Cloud data lifecycle phases.	7	15%
Cloud service providers- Overview and history of Amazon Web Services, Introduction to EC2, Instance Types and Uses, Auto scaling Instances, Introduction to Amazon Machine Images (AMI's), key components and types of Ami's, Elastic IPs, Elastic Load Balancing.	7	13%
Cloud storage- Introduction to Amazon S3 Storage, Buckets and Objects in S3, S3 Storage Classes, Amazon EBS (Elastic Block Store),Amazon Database Services: Amazon RDS (Relational Database Service), Amazon DynamoDB, Amazon Redshift etc.	8	12%
AWS Networking Services- Introduction to AWS Networking, Introduction to VPN and VPC	4	9%
AWS CloudTrail and CloudWatch and alarms, Docker, Kubernetes, and Elastic Beanstalk, CloudWatch, AWS Messaging services (SNS, SQS, SES) and Data Security.	6	12%

Legal and Economic Challenges in Data Protection- Data Recovery and protection, Backup Strategies and recovery considerations, Copyright and Intellectual Property Rights (IPR), International Legal Frameworks.	8	12%
Budget and Cost Management- Budgeting, Resource planning, Introduction to ML ecosystem, Amazon Comprehend, Amazon Forecast, Amazon Fraud Detector, Amazon Kendra, Amazon Lex, Amazon Sage Maker, Amazon Textract, Amazon Transcribe, Amazon Translate.	12	12%
End Term 100% Syllabus		

This document is approved by

Designation	**Name**	**Signature**
Course Coordinator	Dr. Priyanka Gupta	
Head Academic Delivery	Dr. Susheela Hooda	
Dean	Dr. Rupali Gill	
Date (DD/MM/YYYY)	24/06/2025	